Jingzhe Shi

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Education Background

Undergraduate, Yao Class, Tsinghua University **Sep. 2021 – Jun. 2025**
Institute for Interdisciplinary Information Sciences, Tsinghua Univ.; GPA: 3.86/4.0 (overall); 3.90/4.0 (specialized).

- Selected AI-related Courses (4.0/4.0): Machine Learning, Natural Language Processing, Advanced Computer Graphics, Design Thinking in the Age of Artificial Intelligence, Computer Architecture, Operating Systems and Distributed Systems, Mathematics for Computer Science, Theory of Computation, The Physics of Information, Research Immersion Training.

Selected Publications and Preprints

(**Please refer to Google Scholar: https://scholar.google.com/citations?user=x_4IcIoAAAAJ for a full list.**)
(* for equal contribution, # for equal correspondence)

Intrinsic Entropy of Context Length Scaling in LLMs **May 2025**
Jingzhe Shi*, *Qinwei Ma**, *Hongyi Liu**, *Hang Zhao#*, *Jeng-Neng Hwang*, *Lei Li#*
- Accepted by **ICLR 2026** as paper with **Oral Presentation**.
- OpenReview: https://openreview.net/forum?id=vnipyA8c9V.

PRISM-Physics: Causal DAG-Based Process Evaluation for Physics Reasoning **Oct. 2025**
*Wanjia Zhao**, *Qinwei Ma**, ***Jingzhe Shi****, *Shirley Wu*, *Jiaqi Han*, *Yijia Xiao*, *Si-Yuan Chen*, *Xiao Luo*, *Ludwig Schmidt*, *James Zou*
- Accepted by **ICLR 2026**.
- OpenReview: https://openreview.net/forum?id=4PZMeopXzP.

Physics Supernova: AI Agent Matches Elite Gold Medalists at IPhO 2025 **Sep. 2025**
*Jiahao Qiu**, ***Jingzhe Shi****, *Xinzhe Juan*, *Zelin Zhao*, *Jiayi Geng*, *Shilong Liu*, *Hongru Wang*, *Sanfeng Wu*, *Mengdi Wang*
- Accepted by **NeurIPS 2025 LLM Eval** (oral) and **NeurIPS 2025 LAW** (spotlight) workshops.
- ArXiv: https://arxiv.org/abs/2509.01659.

Scaling Law for Time Series Forecasting **May 2024**
Jingzhe Shi*, *Qinwei Ma**, *Huan Ma*, *Lei Li*
- Accepted by **NeurIPS 2024**.
- OpenReview: https://openreview.net/forum?id=Cr2jEHJB9q.

CHOPS: CHat with custOmer Profile Systems for Customer Service with LLMs **Apr. 2024**
Jingzhe Shi, *Jialuo Li*, *Qinwei Ma*, *Zaiwen Yang*, *Huan Ma*, *Lei Li*
- Accepted by **COLM 2024**.
- OpenReview: https://openreview.net/forum?id=9Wmdk94oKF.

Large Trajectory Models are Scalable Motion Predictors and Planners **Oct. 2023**
Qiao Sun, *Shiduo Zhang*, *Danjiao Ma*, ***Jingzhe Shi***, *Derun Li*, *Simian Luo*, *Yu Wang*, *Ningyi Xu*, *Guangzhi Cao*, *Hang Zhao*
- Preprint.
- ArXiv: https://arxiv.org/abs/2310.19620.

Honors and Awards

Gold Medal in IPhO 2021 (top 10 world-wide) **Jul. 2021**
International Physics Olympiad (IPhO)
- Gold Medal, ranking 10th globally in the 51st International Physics Olympiad (IPhO 2021).

National Team Member for Physics Olympaid (top 5 nation-wide) and Gold Medal in CPhO **Sep. 2020 - Jan. 2021**
Chinese Physics Olympiad (CPhO)
- One of the top 5 students out of 660k contestants selected to represent China mainland to attend International Physics Olympiad (i.e. IPhO) through a series of Chinese Physics Olympiads (i.e. CPhO). In this process I won a series of awards including a Gold Medal in CPhO.

First-Class Freshmen Scholarship **Nov. 2021**
Tsinghua University
- Top scholarship for freshmen at Tsinghua University ($< 10/89$).

Technological Innovation Scholarship **Nov. 2023 & Nov. 2024**
Tsinghua University
- Scholarship for innovative research or technological outcomes at Tsinghua University ($< 10/89$).

Excellent Voluntary & Public Service Scholarship **Nov. 2024**
Tsinghua University
- Scholarship for excellent voluntary or public service at Tsinghua University ($< 10/89$).

Selected Social Work Experience

CPHOS
Co-founder, former tech group leader, council member *Dec. 2020 – Present*
- CPHOS (https://cphos.cn) is an academic non-profit organization providing Physics Olympiad simulations 4–5 times per year for high school contestants, free of charge, via an online platform.
- Founded in late 2020 by a group of 10 (including myself), now with 100+ members from top universities in China; 1000+ students from 200+ high schools participate in most Olympiads organized by CPHOS.
- Led the tech group to develop tools supporting online Olympiads, including an LLM-based replier for customer service (**which developed into the CHOPS research project accepted by COLM**).

Optiver, Shanghai Office
Trading Intern *Jul. 2024 – Aug. 2024*
- Optiver (https://optiver.com) is a leading global market-making firm.

Service

- **Reviewer**: ICLR 2025 & 2026, COLM 2025 & 2026, NeurIPS 2025.
- **IPhO 2022 Marker** — Invited online marker for the 52nd International Physics Olympiad (IPhO 2022).
  In 2022, IPhO was held in Switzerland and, due to the pandemic, extra markers were invited. I was invited and served as an online marker: grading, coordinating with my marker partner, and conducting rebuttals with team leaders from around the world via online meetings.

Skills and Others

- **Languages:** Chinese (native); English (TOEFL: 113; R30, L28, S26, W29; test taken in Nov. 2024); Japanese (daily dialogue).
- **Programming languages:** Python, C/C++, etc.
- **Tools:** Git, LaTeX, SQL, etc.